Understanding the Laplace Transform

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1 Introduction

The Laplace transform of a function $f(t)$ is denoted as $\mathcal{L}\{f(t)\}$ or simply $F(s)$.

2 Definition

The Laplace transform of a function $f(t)$ is defined as in Equation 1.

$$\begin{aligned} F(s) &= \mathcal{L}\{f(t)\} \\ &= \int_0^\infty e^{-st} f(t) dt \end{aligned} \tag{1}$$

Here: $F(s)$ is the Laplace transform of the function $f(t)$, s is the complex frequency parameter, $f(t)$ is the time-domain function.

3 Inverse

The inverse Laplace transform allows us to recover the original function $f(t)$ from its Laplace transform $F(s)$ as depicted in Equation 2.

$$f(t) = \frac{1}{2\pi i} \int_{e-i\infty}^{e+i\infty} e^{st} F(s) ds \tag{2}$$

The integration path in the complex plane is typically chosen so that it includes all the singularities of $F(s)$.

4 Properties

4.1 Linearity

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s) \tag{3}$$

4.2 Time-shifting

$$\mathcal{L}\{e^{at} f(t)\} = F(s - a) \tag{4}$$

4.3 Frequency-shifting

$$\mathcal{L}\{f(t-a)\} = e^{-as}F(s) \tag{5}$$

4.4 differentiation

$$\mathcal{L}\{\frac{d^n}{dt^n}f(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - ... - f^{(n-1)}(0) \tag{6}$$

5 Conclusion

Tasks

1. Replicate at least 1 inline mathematical expressions
2. Replicate Equation 1,2 and 6
3. Replicate the references to the numbered equations